

Data Mining 1 Project

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1 DATA UNDERSTANDING AND PREPARATION

The goal of the project is to analyze the IMDb Dataset, which contains data about movies, TV shows, and other forms of visual entertainment, along with their ratings. Each record includes key information about the title, as well as insights into critical aspects such as awards and reviews, as well as statistical ratings and other metadata. The dataset is updated as of September 1, 2024.

1.1 Data Semantics and First Global Exploration of the Dataset

The IMDb Dataset contains 16431 records and 23 attributes, of which 10 are categorical and 13 numeric. They are listed in Tables 1 and 2. Among the **numerical attributes**, most of them are discrete and ratio-scaled, as they represent counts, with *startYear* and *endYear* being interval values. Only *runtimeMinutes* can be classified as a continuous attribute, as it represents a time duration, even though it only assumes integer values in the dataset. Among the **categorical attributes**, *canHaveEpisodes*, *isRatable* and *isAdult* are **binary**, *rating*, *worstRating* and *bestRating* are **ordinal**, while the rest are all **nominal**.

Looking at the records' data types, we can observe some syntactic inaccuracies: *isAdult* should be converted from integer to boolean, whereas *rating* assumes ten categorical values of the form '(0, 1]', '(1, 2]', ..., '(9, 10]'. We can assume they represent the binning of a continuous rating, therefore we map them into an integer space [1,10] to ease later analysis. Just by looking at the head of the dataframe, we can also tell that *countryOfOrigin* and *genres* contain collections of categorical values; therefore, we transform them into lists to allow for easier handling. Some discrete numerical attributes (*endYear*, *awardWins*) are stored as float rather than boolean because `int64` does not support NaN values; however, we do not deem it necessary to convert them.

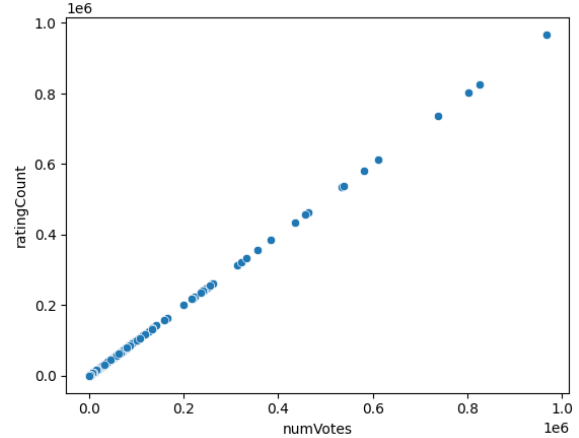


Figure 1: Scatterplot ratingCount~numVotes.

A first global analysis of the dataframe's summary and descriptive statistics already gives us valuable insight into which variables have missing values¹, as reported in Table 3, and into which features are uninformative: *bestRating*, *worstRating*, and *isRatable* can be safely dropped, as they only assume one value for all records (10, 1, and 1, respectively). We can also see that *numVotes* and *ratingCount* share very similar, when not identical, statistics. The scatterplot in Figure 1 proves their correlation close to 1, which leads us to drop *ratingCount* and only keep *numVotes*.

1.2 Distribution of categorical variables and statistics

In an effort to analyze the univariate distribution of categorical variables, we produced barplots for *titleType* (Figure 2), *countryOfOrigin* (Figure 3), and *genres* (Figure 4). All distributions are imbalanced, being dominated by few very common classes: *movie* and *tvepisode*, the *US*, *drama* and *comedy*, respectively. A barplot for *rating*², in Figure 5,

¹ As the missing values were recorded inconsistently in the dataset, we used the `read_csv` function, passing a list of strings to recognize as missing values to the parameter `na_values`.

² In this data exploration phase, we analyzed this ordinal variable both from a categorical and a numeric point of view, as to maximize insights.

Feature	Description	Type	Pandas dtype
runtimeMinutes	Primary runtime of the title, in minutes.	Continuous. Ratio-scaled.	float64
startYear	Represents the release year of a title. In the case of TV Series, it is the series' start year.	Discrete. Interval.	int64
endYear	TV Series end year.	Discrete. Interval.	float64
ratingCount	The total number of user ratings submitted for the title.	Discrete. Ratio-scaled.	int64
numVotes	Number of votes the title has received.	Discrete. Ratio-scaled.	int64
numRegions	The regions number for this version of the title.	Discrete. Ratio-scaled.	int64
totalImages	Total Number of Images for the title within the IMDb title page.	Discrete. Ratio-scaled.	int64
totalVideos	Total Number of Videos for the title within the IMDb title page.	Discrete. Ratio-scaled.	int64
totalCredits	Total Number of Credits for the title.	Discrete. Ratio-scaled.	int64
criticsReviewTotal	Total Number of Critic Reviews.	Discrete. Ratio-scaled.	int64
awardWins	Number of awards the title won.	Discrete. Ratio-scaled.	float64
awardNominationExclWins	Number of award nominations excluding wins.	Discrete. Ratio-scaled.	int64
userReviewsTotal	Total Number of Users Reviews.	Discrete. Ratio-scaled.	int64

Table 1: Numeric attributes.

Feature	Description	Type	Pandas dtype
originalTitle	Original title, in the original language.	Categorical; mostly unique classes (i.e., names).	object
titleType	The type/format of the title (e.g., movie, short, episode, video, etc.).	Categorical, 10 classes.	object
countryOfOrigin	The country where the title was primarily produced. Some titles can belong to more than one class.	Categorical, 153 classes. A record can belong to more than one class.	object
genres	The genre(s) associated with the title (e.g., drama, comedy, action). Some titles can belong to more than one class.	Categorical, 29 classes. A record can belong to more than one class.	object
canHaveEpisodes	Whether or not the title can have episodes.	Asymmetric binary.	bool
isRatable	Whether or not the title can be rated by users.	Asymmetric binary.	bool
isAdult	Whether or not the title is for adults.	Asymmetric binary.	int64
rating	IMDb title rating class.	Ordinal, 10 values.	object
worstRating	Worst title rating.	Ordinal, supposedly [1,10].	int64
bestRating	Best title rating.	Ordinal, supposedly [1,10].	int64

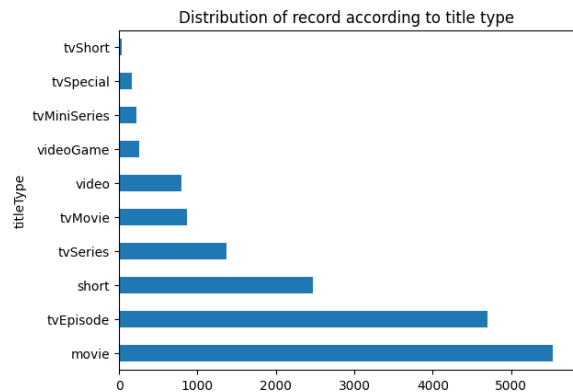
Table 2: Categorical, ordinal, and binary attributes.

Table 3: Features with missing values.

Feature	Non-Null Records
endYear	814
runtimeMinutes	11579
awardWins	13813
genres	16049

shows a left-skewed distribution, with the mode being the [7,8) category.

By looking at the categories for the different variables, we question the informativeness of *canHaveEpisodes* and *isAdult*. The first assumes positive values only for the title types *tvSeries* and *tvMiniSeries*: we keep it, as an explicit feature of serialized content. *isAdult* proves to be redun-

Figure 2: Barplot for *titleType*.

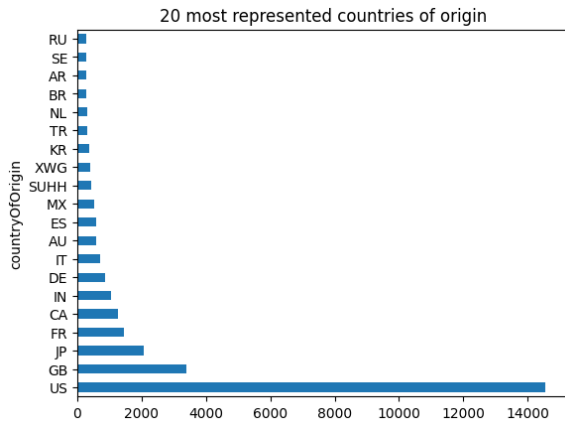


Figure 3: Barplot for *countryOfOrigin* (top 20).

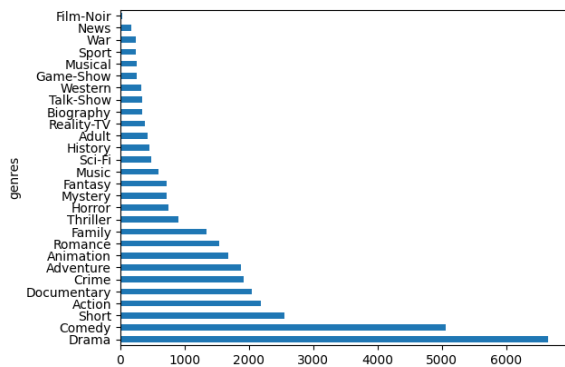


Figure 4: Barplot for *genres*.

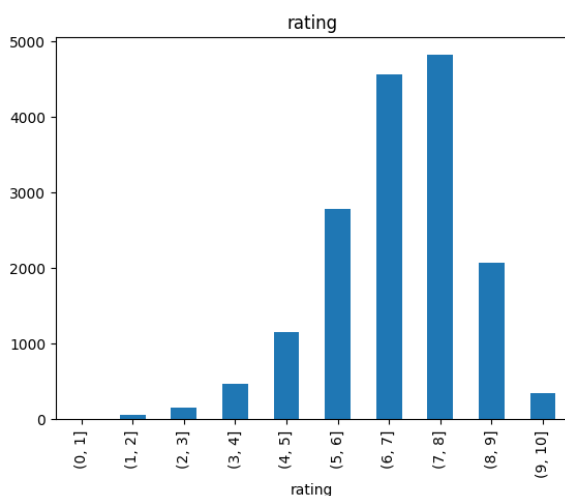


Figure 5: Barplot for *rating*.

dant as its information is conveyed by the *adult* category in *genres*, therefore we drop it.

1.3 Distribution of numeric variables and statistics; transformation of variables

Before analyzing numeric variables, we add the following features:

- *moreCountriesOfOrigin*: the number of countries listed for the record for the variable *countryOfOrigin*. For the training set, the values range from 1 to 10.
- *numGenres*: the number of genres listed in *genres*. For the training set, the values range from 1 to 3.
- *reviewsTotal*: the sum of *criticReviewsTotal* and *userReviewsTotal*.
- *criticReviewsRatio*: the proportion of review from critics (*criticsReviewTotal*) on the total number of reviews of a record (*reviewsTotal*). For unreviewed records, it is set to zero.
- *awardsAndNominations*: the sum of *awardWins* and *awardNominationsExcludeWins*.

Moreover, we use the available interval features, *startYear* and *endYear* to generate the timeline graph in Figure 6. We can observe how the number of titles being released each year has consistently grown until the end of the 2010s, with two noticeable dips in this trend: around 2001 (which could be linked to the aftermath of 9/11) and after 2020 (probably due to the Covid-19 pandemic). The time series for *endYear* follows the same trend on a smaller scale: we suspect a positive correlation between the two variables. There are no end dates available before the 1940s, which is understandable, as serialized content was not produced until around the 1930s.

In order to have a first global view of both the univariate distribution of the numeric variables singularly, and of possible interesting pairwise correlations, we build a pairplot with a KDE graph for each variable in the diagonal. Our previous intuition about a positive correlation between *startYear* and *endYear* is proved by the scatterplot in Figure 7, therefore we drop *endYear*, which is more problematic due to the large number of missing values.

Most numeric features (*awardWins*, *numVotes*, *totalImages*, *totalVideos*, *totalCredits*, *criticReviewsTotal*, *awardNominationsExcludeWins*, *awardsAndNominations*, *numRegions*, *userReviewsTotal*, *num-*

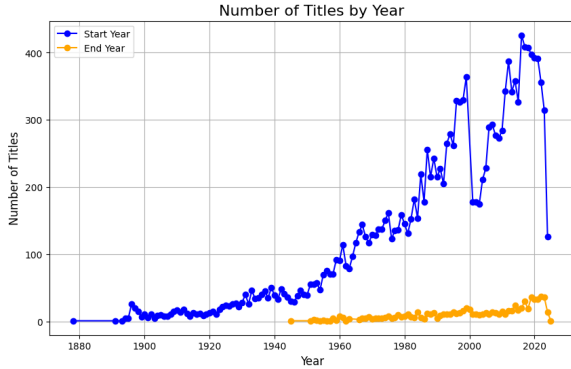


Figure 6: Time series: number of titles being released and ended by year.

CountriesOfOrigin, *reviewsTotal*) have heavily right-skewed distributions. It makes sense that the distribution of these types of features would approximate a power law. We can address this problem using a log transformation of these features.

On the other hand, *runtimeMinutes* also has a right-skewed distribution, but it really doesn't make sense for it to approximate a power law. When excluding outliers (values higher than 3rd quartile + 1.5 IQR), we can see that, for the most part, runtimes are dependent on the title type and that the distribution of the runtime for each title type approximates a bell curve (Figure 8); therefore, we prefer not to log transform this variable and be mindful of the presence of outliers. Examining the titles with a longer runtime reveals that part of the outliers is due to the inconsistent strategy used for the computation of the runtime for serialized titles (series and miniseries): in some cases, the episode runtime is provided, in others, it reports the runtime for the entire series, i.e. the sum of all its episodes' runtimes (some examples are reported in Table 4).

originalTitle	runtimeMinutes	titleType
Alim Day1	3000	tvSeries
Jerry Lewis MDA Labor Day Telethon	1290	tvSeries
Voice of the Planet	600	tvMiniSeries
Orbius	570	movie
Heritage: Civilization and the Jews	540	tvSeries

Table 4: Top 5 titles with the longest runtime.

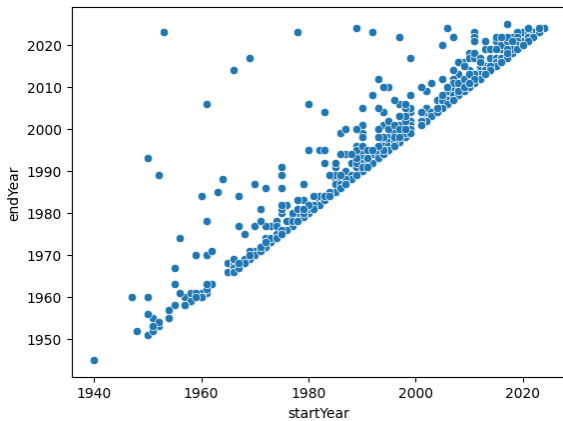


Figure 7: Scatterplot endYear~startYear.

1.4 Handling Missing Values

As explained in Subsection 1.1 and summarized in Table 3, four variables in the dataset have been found to contain missing values: *endYear*, *runtimeMinutes*, *awardWins*, *genres*.

In Subsection 1.3, we opted to drop *endYear* because its value was missing for most records. This decision is corroborated by the fact that, on one side, it heavily correlates with *startYear*, and on the other, the only records that do have an end year are of the type *tvSeries* and *tvMiniSeries*, and the information about the serial nature of media is already carried by the feature *canHaveEpisodes*.

awardWins has almost three thousand missing values. Given its highly unbalanced distribution, with the large majority of the records having no awards, it makes sense to fill them with the most frequent value, i.e. zero.

The stratified barplot in Figure 9 shows that there is some variation in the frequency of differ-

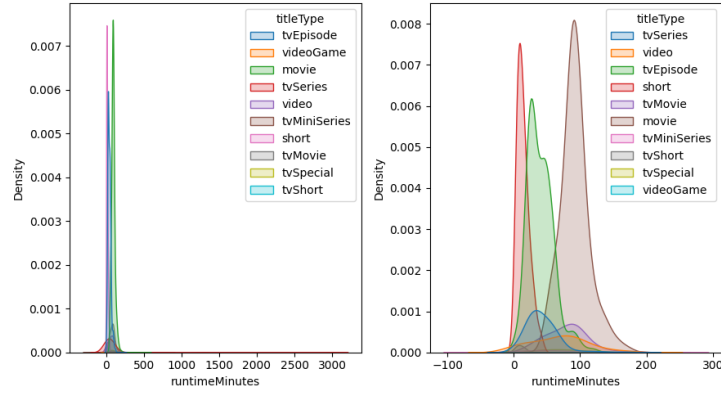


Figure 8: KDE for runtimeMinutes per title type, including and excluding outliers.

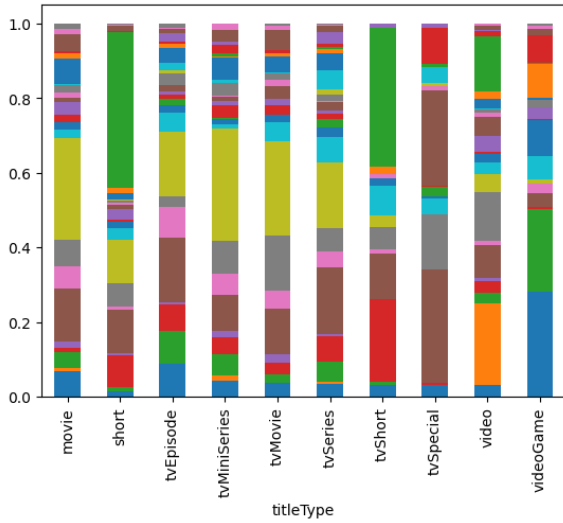


Figure 9: Title type frequency grouped by genre.

ent genres with respect to *titleType*. We decided to fill the 382 missing values for *genres* with the most frequent list of genres given the title type of the record, as shown in Table 5.

titleType	top	# NaN
movie	[Drama]	230
tvSeries	[Comedy]	53
tvMovie	[Drama]	25
video	[Adult]	23
tvSpecial	[Comedy]	18
tvMiniSeries	[Drama]	15
videoGame	[Action, Adventure, Fantasy]	12
tvEpisode	[Comedy]	6

Table 5: Top genre for each title type with missing values for *genres*.

As previously mentioned in Subsection 1.3, *runtimeMinutes* is dependent on the *titleType* of a record (cf. Figure 8): movies are generally longer than TV episodes, and shorts are typically shorter than both. Since we have some missing values for

runtimeMinutes, we can use the median value for the respective type to fill them (Table 6).

titleType	median runtime
movie	90
short	12
tvEpisode	40
tvMiniSeries	60
tvMovie	86
tvSeries	31
tvShort	10
tvSpecial	60
video	76
videoGame	28

Table 6: Median runtime (in minutes) by title type.

1.5 Pairwise correlations and eventual elimination of variables

We observed pairwise linear correlation among numerical variables (after operating the log transformation detailed in Subsection 1.3) by computing their correlation heatmap, displaying Pearson’s correlation coefficient for each pair of features (Figure 10).

numVotes, *userReviewsTotal*, *criticReviewsTotal* and *reviewsTotal* are all highly positively correlated with each other (with a correlation coefficient greater than 70). We only keep *numVotes*, which has the least skewed distribution.

Since the distribution of awards and nominations is not simply skewed, but the majority of records received neither, we binarize *awardsAndNominations* (Table 7) and drop both *awardWins* and *awardNominationsExcludeWins*.

Again, the majority of records do not have videos: we drop *totalVideos* and substitute it with a boolean feature, *hasVideos* (Table 8).

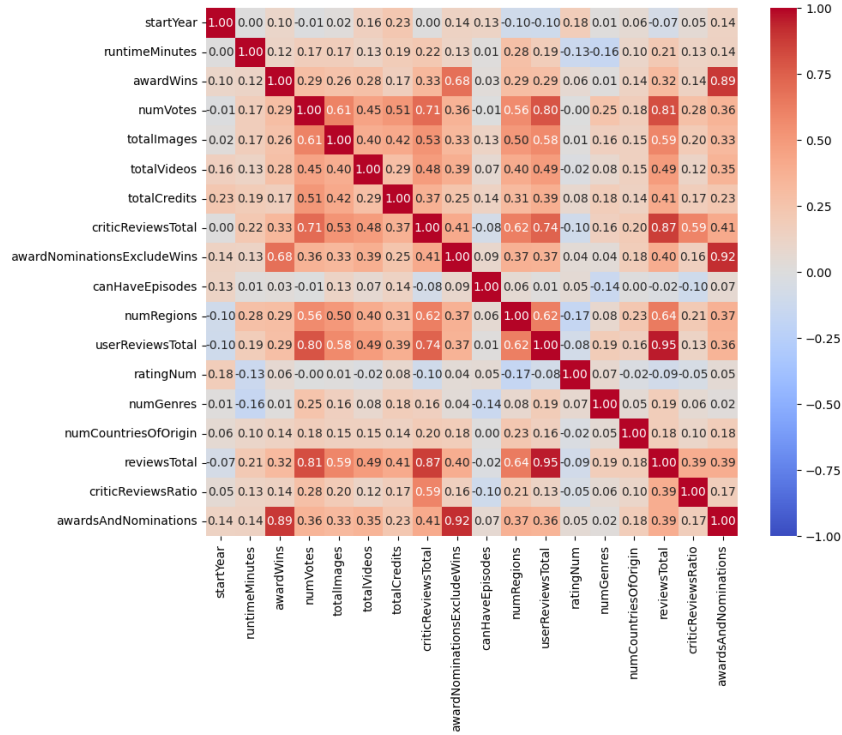


Figure 10: Correlation heatmap before attribute removal.

awardsAndNominations	
False	13692
True	2739

Table 7: *awardsAndNominations* boolean values counts.

hasVideos	
False	14821
True	1610

Table 8: *hasVideos* boolean values counts.

Similarly, most records only have one country of origin, thus we substitute *numCountriesOfOrigin* with the boolean *moreCountriesOfOrigin* (Table 9).

moreCountriesOfOrigin	
False	15285
True	1146

Table 9: *moreCountriesOfOrigin* boolean values counts.

The overall number of features was reduced from 23 to 18; they are listed in Tables 10 and 11. Among the remaining numeric features, we observe in the heatmap in Figure 11 that *numVotes* positively correlates to *totalImages* (0.61, moderate to high correlation), to *numRegions* (0.56, moderate correlation), and to *totalCredits* (0.51, low-moderate correlation).

2 CLUSTERING

2.1 Centroid-based Clustering

In analyzing the dataset using centroid-based methods, we experimented with two different algorithms, K-Means and Bisecting K-Means. For each experiment, we followed the same steps:

1. Choosing the attributes and scaling the dataset: we used all numeric features (cf. Table 10), log-transformed if necessary as detailed in Subsec. 1.3, and scaled the values using *StandardScaler*.
2. Selecting the best value for k : we used both the Elbow method and the Silhouette method as heuristics to determine the appropriate number of clusters. To do so, we iteratively run the algorithm for k in the range $[2, 50]$, plotting the SSE and Silhouette score for each k and comparing the two curves to find the best trade-off.
3. Fitting the algorithm and evaluating the resulting clustering in terms of cohesion and separation by calculating the SSE, Silhouette score, and correlation between the distance matrix of the dataset and the ideal similarity matrix.

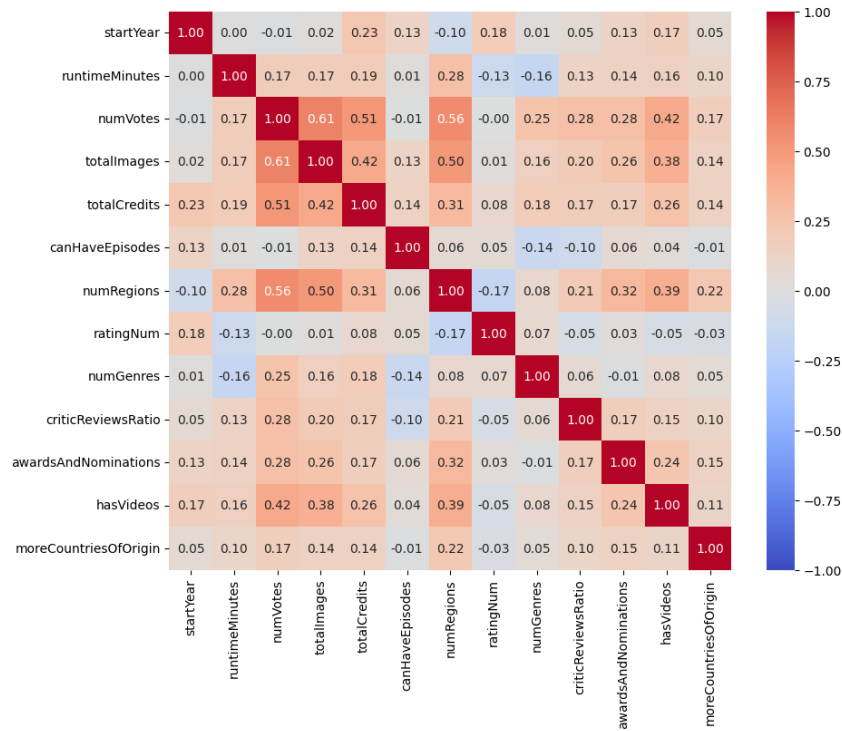


Figure 11: Correlation heatmap after attribute removal.

- Visual cluster exploration: cluster visualization through PCA, t-SNE, and a parallel coordinates plot of the centroids, followed by a further visualization of the clusters' composition w.r.t. categorical and numeric features.

2.1.1 K-Means

We select $k=7$ as a trade-off between SSE and Silhouette score. The resulting SSE and Silhouette score are 62588.082 and 0.189, respectively, while the correlation is -0.458. Though the SSE value might seem high, it is significantly lower than the SSE of 500 randomly generated datasets (cf. Figure 12), which suggests that our clustering is meaningful and not a product of chance.

The clusters' composition is reported in Table 12: in terms of population, we can distinguish Cluster 0 as the largest cluster, with almost 5000 records, followed by Cluster 2 with over 3000 records. The smallest cluster is Cluster 6, with only 950 records. The remaining clusters have between 1400 and 2000 records.

The first thing that emerges from the parallel coordinates graph (Figure 13) is how Cluster 3 is well-separated from the others w.r.t half of the features: it has a particularly high value for *numVotes*, *totalImages*, *totalCredits*, and *numRegions* – variables for which we had observed moderate correlation in Subsec. 1.5. Cluster 5 has a sig-

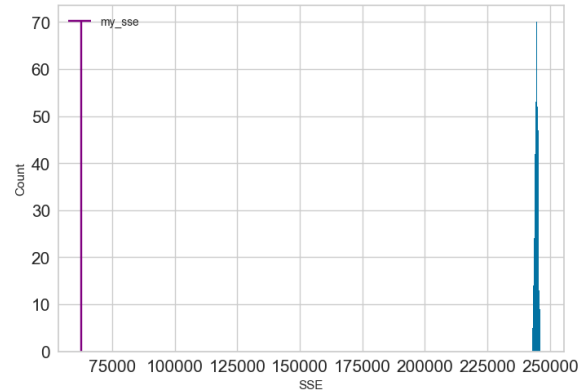


Figure 12: Statistical framework for clusters' validity: see (K-Means).

nificantly higher value for *criticReviewsRatio*: the algorithm might have isolated titles with greater critical appeal. Clusters 6 and 4 have the lowest runtime by far; Cluster 6, the smallest one, is also characterized by a low value for *startYear* and *totalCredits*: it probably contains the oldest and shortest titles. Clusters 0 and 2 appear to be the least-well separated from the others; they are indeed the biggest clusters.

In order to evaluate the results, we first used stratified bar charts to visualize the distribution of the clusters against known categorical features.

Feature	Description	Type
runtimeMinutes	Primary runtime of the title, in minutes.	Continuous. Ratio-scaled.
startYear	Represents the release year of a title. In the case of TV Series, it is the series' start year.	Discrete. Interval.
numVotes	Number of votes the title has received.	Discrete. Ratio-scaled. Log-transformed.
numRegions	The regions number for this version of the title.	Discrete. Ratio-scaled. Log-transformed.
totalImages	Total number of images for the title within the IMDb title page.	Discrete. Ratio-scaled. Log-transformed.
totalCredits	Total number of credits for the title.	Discrete. Ratio-scaled. Log-transformed.
criticReviewsRatio	The proportion of reviews from critics on the total number of reviews of a record. For unreviewed records, it is set to 0.	Continuous. Ratio-scaled.
awardsAndNominations	Total number of awards and nominations.	Discrete. Ratio-scaled. Log-transformed.
numGenres	The number of genres listed for the variable <i>genres</i> .	Discrete. Ratio-scaled.

Table 10: Final numeric attributes.

Feature	Description	Type
originalTitle	Original title, in the original language.	Categorical; mostly unique classes (i.e. names).
titleType	The type/format of the title (e.g. movie, short, tvseries, tvepisode, video, etc.).	Categorical, 10 classes.
countryOfOrigin	The country where the title was primarily produced.	Categorical, 153 classes. Some titles can belong to more than 1 class.
genres	The genre(s) associated with the title (e.g., drama, comedy, action).	Categorical, 29 classes. Some titles can belong to more than 1 class.
rating	IMDb title rating class.	Ordinal, 10 values.
ratingNum	Mapping of <i>rating</i> to the integers [1,10].	Ordinal, 10 values.
canHaveEpisodes	Whether or not the title can have episodes.	Asymmetric binary.
hasVideos	Whether or not the title IMDb title page has at least one video.	Asymmetric binary.
moreCountriesOfOrigin	Whether or not the title was produced in more than one country.	Binary.

Table 11: Final categorical, ordinal, and binary attributes.

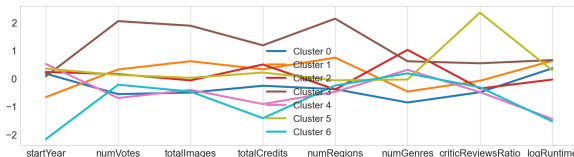


Figure 13: Centroids' visualization for K-Means.

Cluster	# of records
0	4841
1	1985
2	3424
3	1424
4	1985
5	1822
6	950

Table 12: Clusters for K-Means.

Cluster	# of records
0	1881
1	1398
2	1397
3	1712
4	3595
5	4184
6	2264

Table 13: Clusters for BK-Means.

- *titleType* (cf. the heatmap in Figure 14): the smallest cluster, Cluster 6, is the purest, being mainly made up of shorts. Nevertheless, it's Cluster 4 that contains the largest number of shorts (as well as some tvEpisodes). It also stands out for having the fewest movies – a class otherwise well represented across all other clusters. Cluster 2 has a very large proportion of tvEpisode, which explains its centroid's medium-low position for runtime in the parallel coordinates plot. The importance of *logRuntime* for explaining this clustering is also highlighted by the fact that the few tvShorts present have been almost en-

tirely distributed amongst these three clusters. Cluster 0, the biggest, contains a great number of almost all title types.

- Boolean variables: for almost all clusters, the True records represent a small proportion, except for Cluster 3: around half of its records are True for *hasVideos* and *awardsAndNominations*. Given the insights from the parallel coordinates plot, we can assume this cluster contains mainstream, popular content: it has

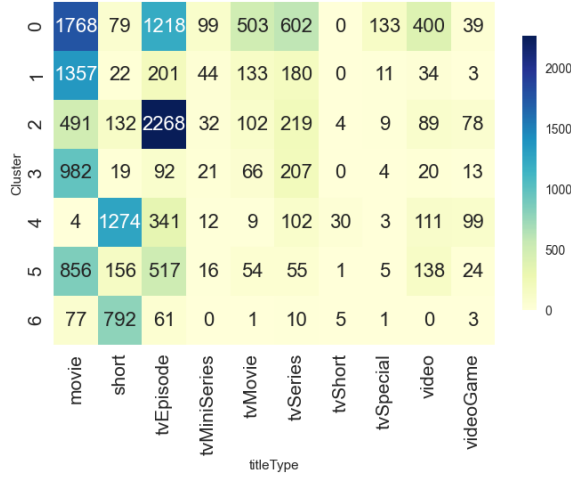


Figure 14: Heatmap Clusters vs titleType for K-Means.

relatively more videos, images, and votes on its page and was distributed in more countries, but it doesn't have the highest ratio of critical reviews. Cluster 6, on the other hand, has little to no True values for these variables. The highest proportion of True values for *canHaveEpisodes* is in Cluster 0, which contains the largest amount of tvSeries and tvMiniSeries.

We also analyzed the clusters' composition w.r.t. the numeric variables used by the algorithm, expanding the insights gained from the parallel coordinates plot. Interestingly, the two purest clusters w.r.t. to *numGenres* are the biggest: Cluster 0 mostly has one genre, with a minority of two genres, while Cluster 2 mostly has three genres, with a small proportion of two genres. The other clusters are more heterogeneous, having all three possible values. The line graph in Figure 15 shows the distribution of the clusters w.r.t. *startYear*: as hinted by its centroid, Cluster 6 contains the majority of records released before 1930s, and has no record released after 1983. On the other hand, Cluster 4 only has records released since 1963: the algorithm has therefore grouped older shorts in a cluster and newer shorts in another. Cluster 1, which also had a relatively low centroid, mostly contains records from the 1930s until the 1990s. Boxplots for the remaining variables also confirm the insights that were previously discussed. For *runtimeMinutes*, Clusters 4 and 6 have the lowest median, at 12 minutes, as well as a short box, indicating low variability. Cluster 4 follows at 40 minutes, coherently with the length of a tvEpisode.

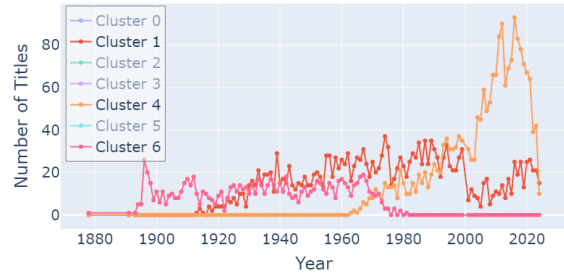


Figure 15: Number of titles by year for relevant clusters for K-Means.

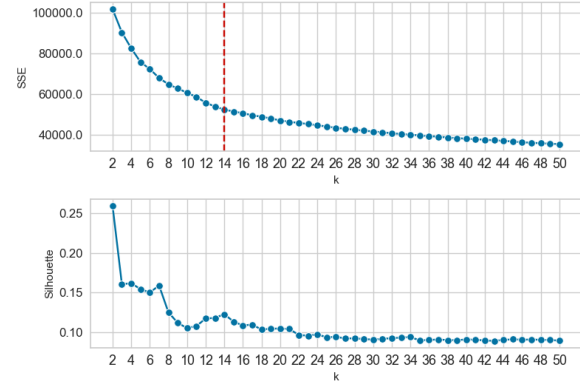


Figure 16: SSE and Silhouette score for choosing k (BK-Means).

2.1.2 Bisecting K-Means

We select $k=7$, the only significant local peak in the Silhouette plot (Figure 16). All evaluation metrics are worse than the ones for K-Means, with a SSE equal to 67857.023, a Silhouette score equal to 0.128, and a correlation of -0.393. Again, we compared the SSE with the one of 500 randomly generated datasets to confirm its meaningfulness. The clusters' sizes are more homogeneous than K-Means' (cf. Table 13): Cluster 5 is the biggest with around 4000 records, followed by Cluster 4 with almost 3600 records. All other clusters range between around 1400 and 2300 records.

Analyzing the parallel coordinates plot in Figure 17, we can see that similarly to before, there's a cluster, Cluster 2, that is well-separated from the others w.r.t. *numvotes*, *totalImages*, *totalCredits* and *numRegions*. In terms of these values, it is followed by Clusters 0 and 1. Moreover, there's once again a cluster, Cluster 1, with a particularly high value for *criticReviewsRatio*. Differently from K-Means, there is one single cluster, Cluster 3, with a significantly lower value for both *logRuntime* and *totalCredits*. It also has the lowest value for *startYear*, together with Cluster 6. Differently from before, it seems that the algorithm has created a



Figure 17: Centroids' visualization for BK-Means.

single cluster for shorts, that probably contains also newer titles. Cluster 4 also has a relatively low value for *logRuntime*; we expect it to contain tvEpisodes and the remaining shorts.

Like for K-Means, we visualized the distribution of the clusters against known categorical features. We find many similarities with the previous algorithm, but also some differences:

- *titleType* (cf. Figure 18): again, the distribution of shorts is what becomes immediately evident. Around half of them are to be found in Cluster 3, which also has a small proportion of tvEpisodes and is the only cluster with almost no movies. Cluster 4, the 2nd biggest cluster, is mostly made of tvEpisodes and a smaller, yet significant, proportion of shorts. The few tvShort recorded are also distributed among these two clusters. It's also worth noting that Cluster 6 contains the biggest number of movies, although being only the 3rd most popular, and that Cluster 2 is the only cluster with almost no tvEpisodes. Cluster 5, the biggest, contains the largest part of the remaining title types (except for videogame, which are mostly in Cluster 4).
- Boolean variables: again, the cluster whose centroid had the highest value for *numVotes*, *totalImages*, *totalCredits* and *numRegions*, Cluster 2, also has the highest proportion of True values for *hasVideos* and *awardsAndNominations*, at around 50%. The biggest cluster has the most True values for *canHaveEpisodes* because it contains most of the series. Cluster 6 has almost no True values for *hasVideos*, probably because it contains older titles.

By visualizing the clusters' composition w.r.t. the numeric variables used by the algorithm, we can see how the line graph for *startYear* (Figure 19) shows important differences to the one for K-Means. Cluster 3 (the shorts' cluster) does contain the majority of records released before 1930s, but it actually has titles from all years: we were correct in assuming that it contains both older and newer shorts. Cluster 6 only has records released between the 1910s until 1999, and contains the majority of records released from 1955 until 1987.

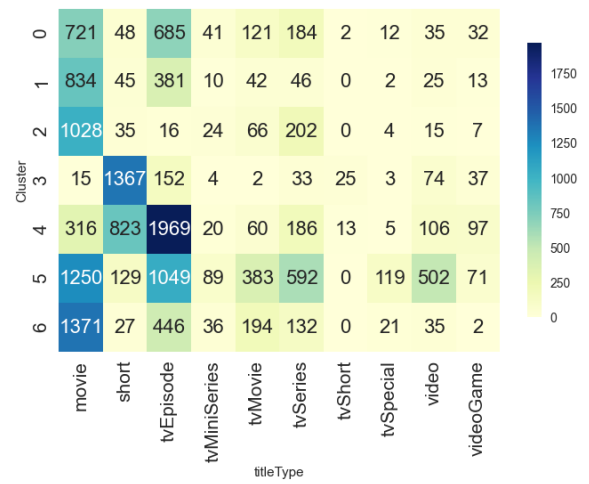


Figure 18: Heatmap Clusters vs titleType (BK-Means).

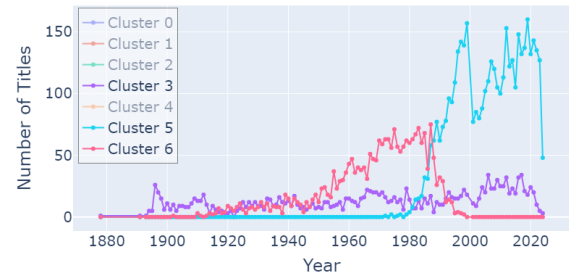


Figure 19: Number of titles by year for relevant clusters for BK-Means.

Cluster 5 (the biggest one) only contains newer titles, released since the 1980s. W.r.t. *numGenres*, only Cluster 4 has no records with one genre: it contains the largest amount of records with three genres, as well as some with two. Clusters 5 and 6 are also quite pure, with a majority of records with one genre, a minority with two, and a negligible amount with three.

In conclusion, although Bisecting K-Means has yielded worse metrics than K-Means, it has still produced meaningful clusters, comparable to those from K-Means. Both algorithms have created one or two clusters containing shorts, one containing a large portion of tvEpisodes, a cluster containing mainstream/popular content, and one with critically interesting titles. Neither clustering is separated w.r.t the rating (for both, the clusters are normally distributed against *ratingNum*).

3 CLASSIFICATION

4 REGRESSION

5 PATTERN MINING